



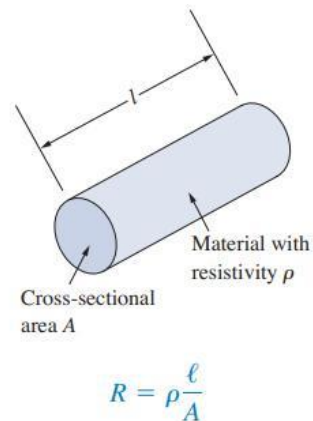
Introduction

Electrical & Electronic Circuits and Elements

Embedded Systems & Embedded Programming

Fundamentals of Electric Circuits (Resistance)

- ❖ Materials have a characteristic behavior of **resisting** the flow of electric charge
- ❖ This behavior is known as **resistance** and its magnitude:
 - Is dictated by electrical properties (**resistivity** - ρ) of the material and its geometry (**A** and **L**)
 - Is represented by **R** and expressed in **ohms** (Ω)
- ❖ The element used to model the current-resisting behavior of a material is known as **resistor**



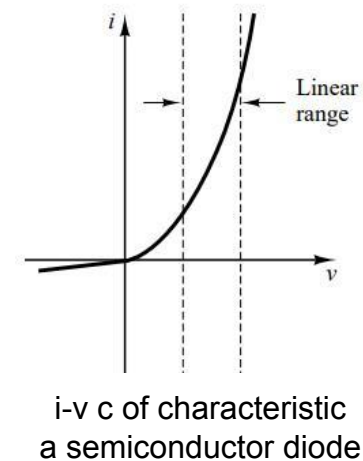
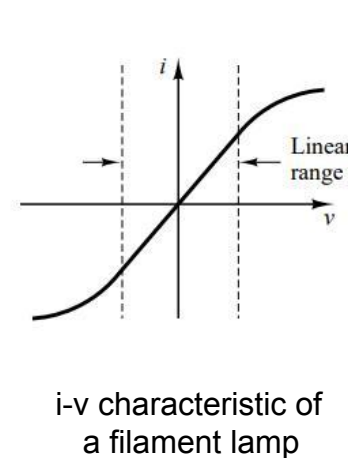
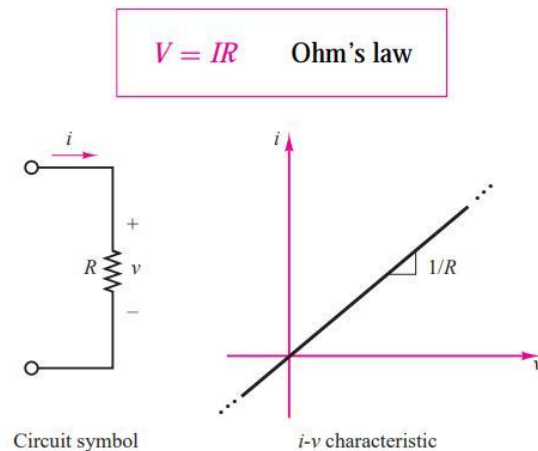
Resistivities of common materials.

Material	Resistivity ($\Omega \cdot m$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

Fundamentals of Electric Circuits (Ohm's Law)

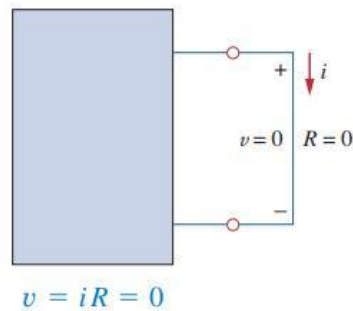
- ❖ The element used to model the current-resisting behavior of a material is known as **resistor**
- ❖ The **voltage** across a resistor is **directly proportional** to the **current** flowing through the **resistor**
 - Ohm's law is an approximation of the resisting behavior of materials
 - Typically, the relationship between current and voltage is not **linear**
 - i-v characteristic graphs can tell us so much about the behavior of elements
 - Resistance, power, passive/active etc.

 [Resistance and Ohm's law](#)

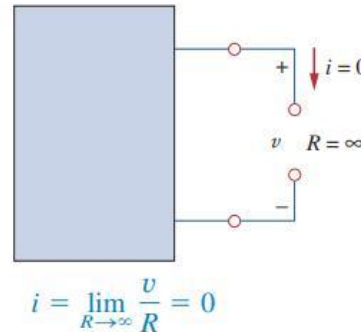


Fundamentals of Electric Circuits (Definitions)

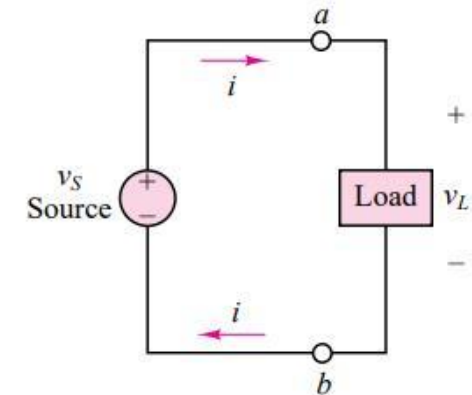
- ❖ **Source:** Any element that provides energy to the circuit
- ❖ **Load:** Any device or circuit powered by electricity
 - Simple; like a lamp, complex; like a modern high-speed computer
- ❖ **Short circuit:** A circuit element with resistance approaching zero
- ❖ **Open circuit:** A circuit element with resistance approaching infinity



Short circuit



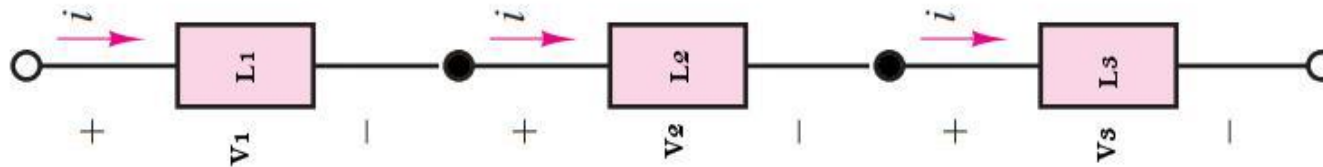
Open circuit



Fundamentals of Electric Circuits (Definitions)

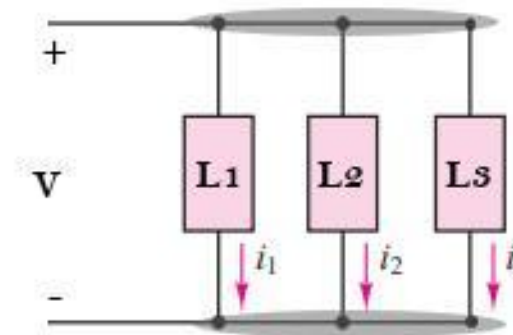
❖ **Series Circuits:** Two or more elements are in series if they

- Exclusively share a single node and
- Consequently carry the same current



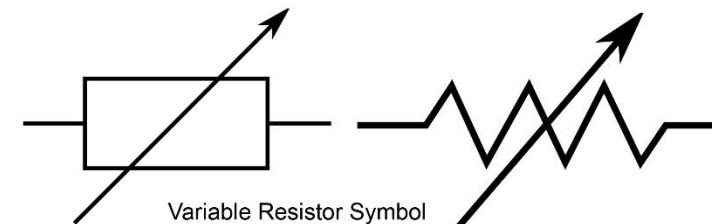
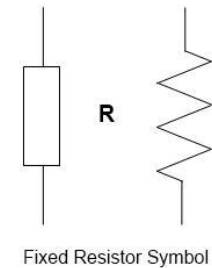
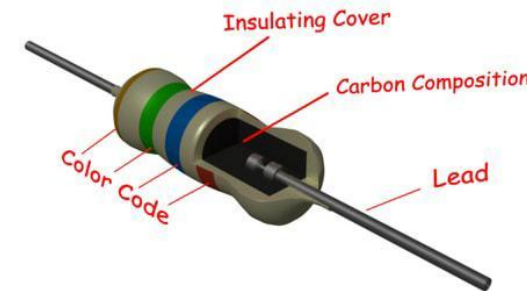
❖ **Parallel Circuits:** Two or more elements are in parallel if they

- Are connected to the same two nodes
- Consequently have the same voltage across them



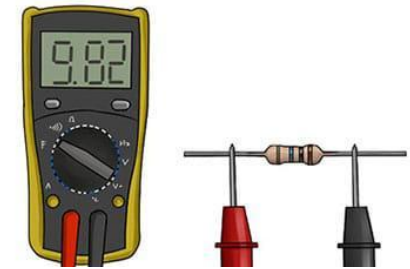
Fundamentals of Electric Circuits (Resistors)

- ❖ **Resistor** is the element used to resist/control/limit electric current flow in circuits
- ❖ It is a passive element and it converts electrical energy to heat energy
- ❖ In a typical resistor, voltage-current relationship is linear ($V = R \times I$)
- ❖ Types
 - Fixed resistors (constant resistance value)
 - Variable resistors (adjustable resistance value)
- ❖ Application
 - Limiting/dividing current in electric circuits
 - Lowering voltage levels (voltage divider)
 - As pull-up and pull-down
 - As sensor



Fundamentals of Electric Circuits (Resistors)

- ❖ Most common resistors are made of either a **carbon**, **metal**, or **metal-oxide** film
- ❖ The resistance of a resistor can be measured by:
 - Ohmmeters (without any power supplied to the resistor)
 - The bands of color on a resistor (4 - 6 bands), [Online Calculator](#)

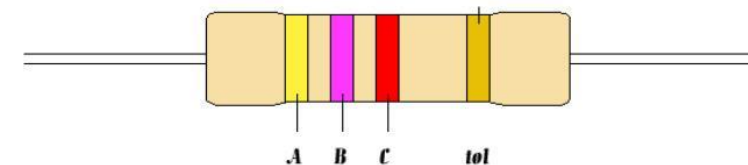


Color	Tolerance
Brown	±1%
Red	±2%
Gold	±5%
Silver	±10%
None	±20%

4 bands color tables

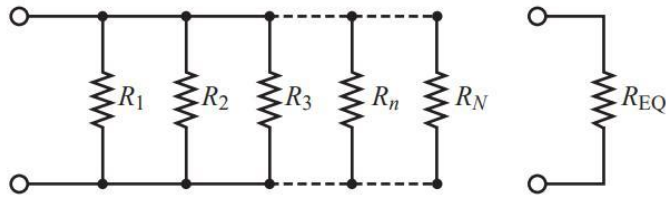
Band color	Digit	Multiplier
Black	0	X1
Brown	1	X10
Red	2	X100
Orange	3	X1000
Yellow	4	X10000
Green	5	X100000
Blue	6	X1000000
Purple	7	X10000000
Grey	8	X100000000
White	9	X1000000000
Silver	-	x.01
Gold	-	x.1

$$\text{Resistor value} = AB \times 10^C \pm \text{tol}\%(\Omega)$$



Fundamentals of Electric Circuits (Resistors)

❖ Resistors in Series and Parallel



$$R_{EQ} = \frac{1}{1/R_1 + 1/R_2 + \cdots + 1/R_N}$$

Equivalent parallel
resistance

$$i_n = \frac{1/R_n}{1/R_1 + 1/R_2 + \cdots + 1/R_n + \cdots + 1/R_N} i_S$$

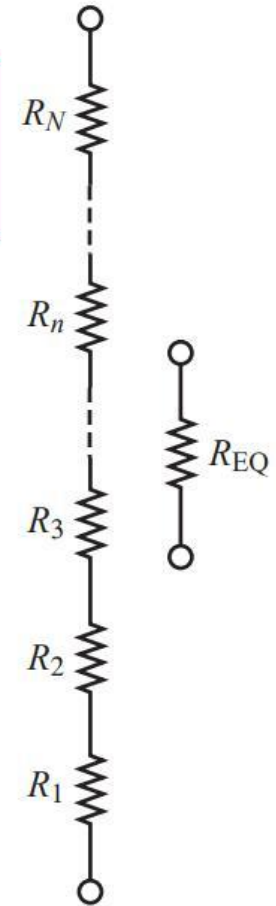
Current
divider

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_n + \cdots + R_N} v_S$$

Voltage divider

$$R_{EQ} = \sum_{n=1}^N R_n$$

Equivalent series resistance



Fundamentals of Electric Circuits (Resistors)

❖ Some examples of variable resistors



A **potentiometer** is a manually adjustable variable resistor with 3 terminals



Varistors have a non-linear resistance which is dependent on the voltage over the varistor. They are used as **over-voltage protection** devices



Photoresistors (LDR) show a decrease in resistance when light intensity increases.



Thermistors are resistors of which the resistance changes significantly when temperature changes



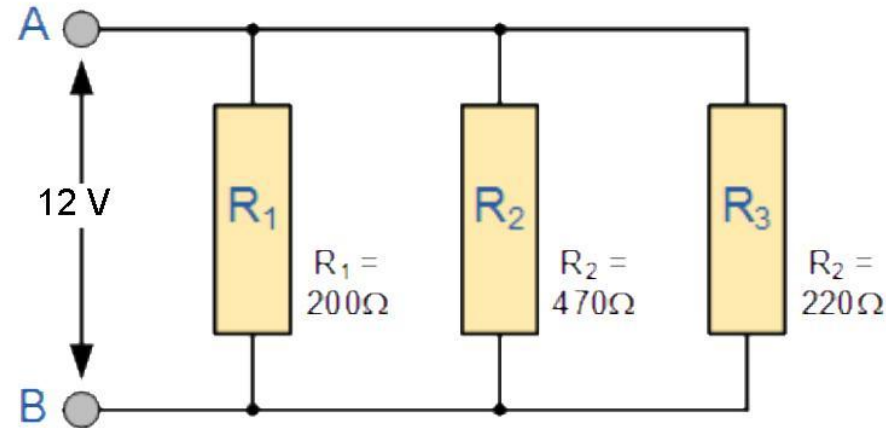
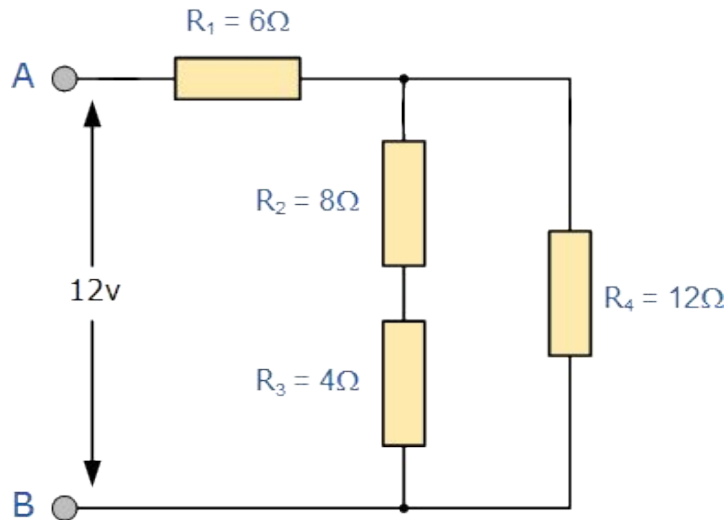
MDR

The **magneto-resistor** type can be used to detect and measure magnetic fields.

[Resistor Types](#)

Fundamentals of Electric Circuits (Resistors)

- ❖ Examples of KVL, KCL and Ohm's law
 - Calculate the current and voltage of every resistor
 - Calculate the power consumed by every circuit



- ❖ Calculate the internal resistance of a 5V battery which can deliver max 10W power

Fundamentals of Electric Circuits (Resistors)

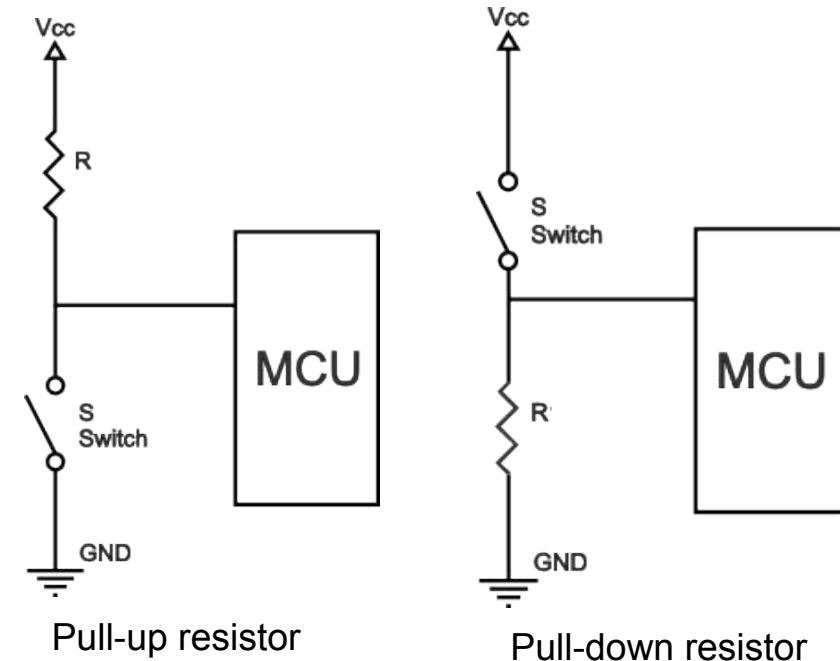
❖ Pull up resistor / Pull down resistor

➤ Logic states of digital circuits

- High (e.g. 2.3 - 3.3 V)
- Low (e.g. 0 - 1.1 V)
- Floating (high impedance)

➤ Resistance value depends on

- Power dissipation
- High/Low of signal levels
- Input leakage current
- Speed of the circuit (RC circuit)
- A rule of thumb: at least 10 times smaller than the impedance of the input pin
- 4.7 k Ω is a common pull up resistor



[Pull Up Resistor Tutorial](#)

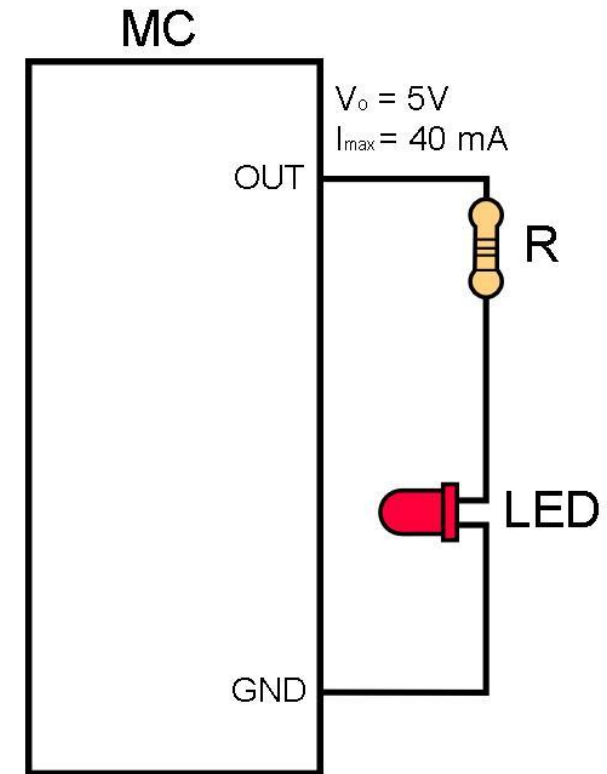
[Calculate pull up/down resistors](#)

Fundamentals of Electric Circuits (Resistors)

❖ LED (Light-Emitting Diode)

- Why do we need a resistor?
 - To limit the current and divide the voltage
- Calculate the resistance of the resistor
 - If the forward current is 17 mA
 - And forward voltage is 2 V

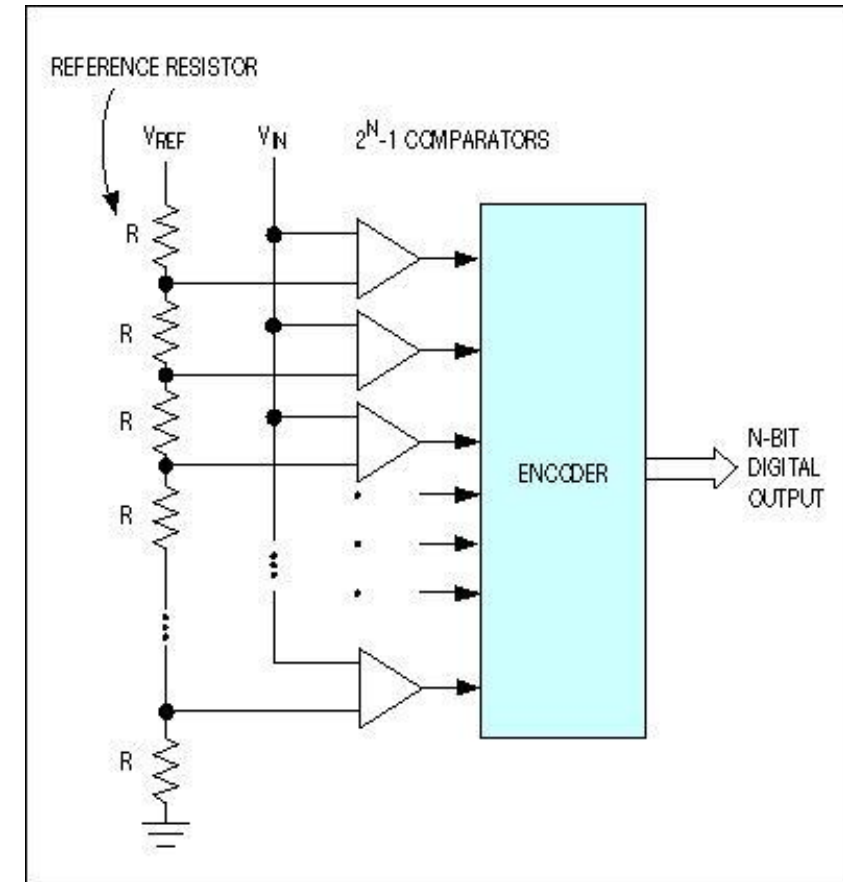
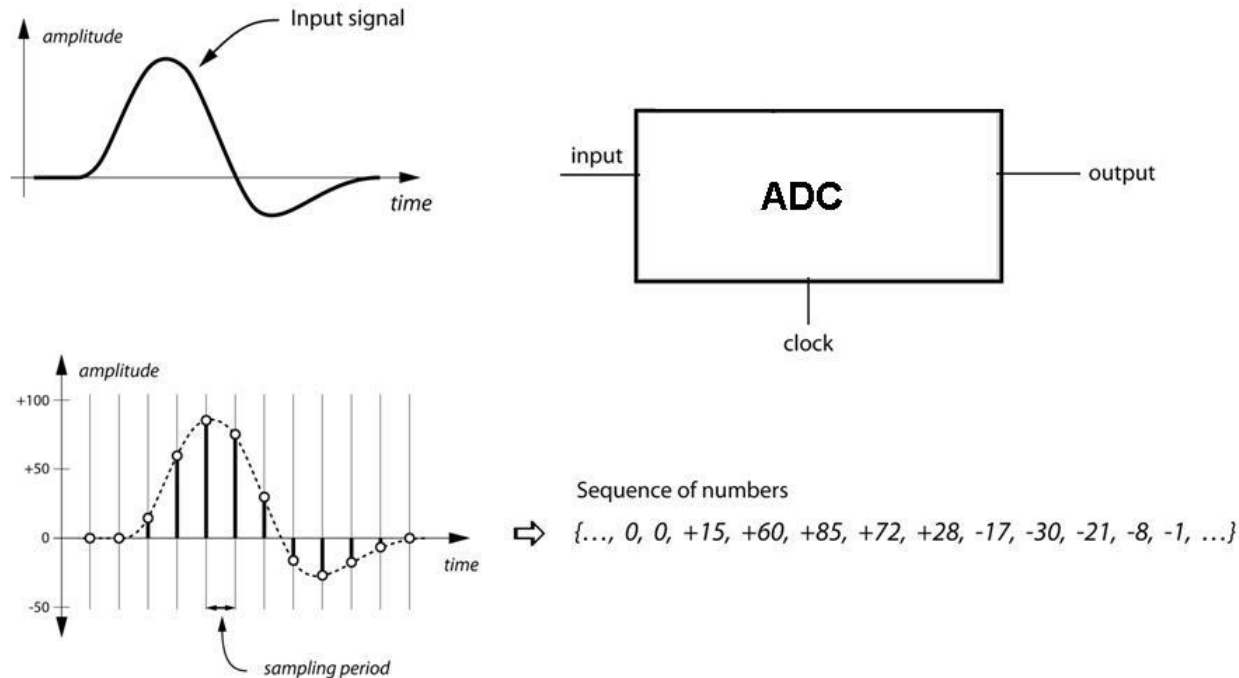
Answer: 176 Ω



[Datasheet of the LED](#)

Fundamentals of Electric Circuits (Resistors)

❖ ADC(Analog to Digital Converter)

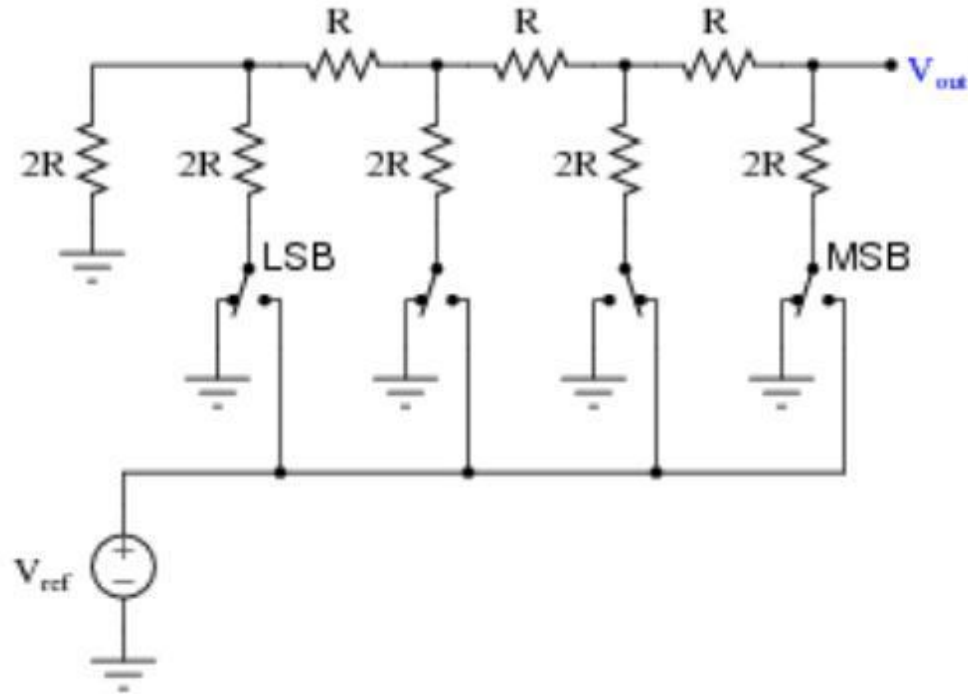


Fundamentals of Electric Circuits (Resistors)

❖ R-2R ladder network **D**igital to **A**nalog **C**onvertor (DAC)

❖ Calculate the out voltage

- If the digital value is 0100
- If the digital value is 1001
- Suppose that V_{ref} is 4V



Fundamentals of Electric Circuits

❖ References

- Fundamentals of Electric Circuits, 5th Edition, Charles K. Alexander and Matthew N.O. Sadiku, ISBN 978-0-07-338057-5
- Fundamentals of Electrical Engineering, First Edition, Giorgio Rizzoni, ISBN 978-0-07-338037-7
- ***Most of the images have been copied from these two books***

❖ Some useful links

- [Resistors - Ohm's Law is not a real law](#)
- [Resistors](#)
- [I-V Characteristic Curves](#)
- [Light-Emitting Diodes \(LEDs\)](#)
- [More about resistors](#)

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a